

TRANSFER FUNCTIONS OF MICROVIBRATIONAL DISTURBANCES ON A SATELLITE

Morio Toyoshima, Takashi Jono, Nobuhiro Takahashi, Toshihiko Yamawaki,
Keizo Nakagawa, and Katsuyoshi Arai
OICETS Project Team, National Space Development Agency of Japan
2-1-1 Sengen, Tsukuba, Ibaraki, 305-8505 Japan
toyoshima.morio@nasda.go.jp

Abstract

Microvibrations from the mechanical moving devices in the Optical Inter-orbit Communications Engineering Test Satellite (OICETS) are measured in a system level test in the ambient environment on the ground by using 40 accelerometers with sensitivity of 1 mG ($1\text{ G}=9.8\text{ m/s}^2$) and dynamic range of 50 dB. The largest source of vibrational disturbance in the satellite is identified in the microvibration test. A simulated OICETS satellite model that consists of a model of the satellite structure and an engineering model of the laser communication terminal is used with some flight components, such as four reaction wheels, two conical-scanning earth sensors, one inertial reference unit, two solar paddle drive motors, one microvibration measurement equipment, and some dummy mass components. The impact of the microvibrations on the optical tracking performance of the onboard laser communication terminal is analyzed with respect to the frequency response based on transfer functions. Optical tracking by the laser communication terminal was maintained even in the worst case condition when all the mechanical disturbing devices in the satellite were turned on. This satisfies adequately the optical tracking performance requirements for intersatellite laser communication links.

1. Introduction

Precision pointing and tracking controls are required to transmit successfully between laser communication terminals in satellite-borne optical communication systems. The optical beamwidth (on the order of microradians) is a few orders of magnitude narrower than the attitude accuracy of the host satellite, which is usually assumed to be about 0.1 degrees (1.75 mrad). The accuracy of the satellite attitude can be improved to better than 0.01 deg (0.175 mrad) by using star trackers.¹⁻³ There are consequent drastic reductions in received signal strength unless special care is taken to reduce the pointing and tracking uncertainty, which is the most serious technological problem of satellite-borne optical communication systems.

Detailed knowledge of satellite microvibrational noise is needed for the proper design and engineering of pointing and tracking systems. The power spectral density (PSD) of the microvibration on the LANDSAT-4 satellite in orbit was first reported by Sudey in 1984,⁴ and the power spectrum of a transient caused by a thruster firing on the OLYMPUS satellite were reported by Wittig in 1990.^{5,6} The microvibration test on the SPOT-4 satellite was performed on the ground and the disturbance forces of a SPOT-4 reaction wheel were reported in 1994.⁵ Vibration signatures in orbit were obtained with a ground-based CO₂ laser radar

while the retractable boom on a satellite was moving.^{7,8} Microvibration tests on the ARTEMIS spacecraft were conducted on the ground, and the in-orbit laser pointing was analyzed using a Monte Carlo simulation in 1997.^{9,10} A real in-orbit measurement of microvibration was reported by using the laser communication equipment onboard the ETS-VI satellite.^{11,12} Vibration effects and possible solutions for satellite optical communication networks were presented to overcome the problems caused by transmitter pointing vibration.¹³ A comparison between the on-ground tests and the in-orbit measurements was made by using the SPOT-4 satellite in 1999 after its launch.¹⁴ The maximum measured acceleration level was about 1 mG near the X-wheel. The noise level generated by the nominal rotation speed of the wheels produced an increase in the environmental noise at low frequencies.¹⁴

In this paper, the characteristics of microvibrations on the Optical Inter-orbit Communications Engineering Test Satellite (OICETS) are reported. The microvibration data from mechanical moving sources in the satellite are measured in three directions. The PSD of each microvibrational source is presented, and the transfer functions from the sources to the mounting point of the laser communication terminal, called the Laser Utilizing Communications Equipment (LUCE), are

calculated using Fourier analysis. The microvibrational level is measured as the rotation speed of the reaction wheel is changed and the time-frequency response of the reaction wheel is analyzed. The damping constant of the damped microvibration along the satellite structural panels is derived experimentally. The influence of the microvibration on the LUCE tracking performance is discussed under the conditions that the vibrational disturbing sources are turned on and off.

2. Resolution requirements for earth observation satellites

Earth observation satellites require high accuracy pointing and stringent line-of-sight stability as well as laser communication terminals. Worldwide high-resolution imagery at the one-to-three meter resolution, which is available commercially by, for example, Space Imaging, Inc., as a result of declassification and deregulation by the Department of Defense of the United States, will generate massive amounts of data, especially if the images are multispectral. Either collected data must be stored and processed on-board the observing satellite in order to reduce the required downlink transmission rates, or higher speed communications links must be used that transmit directly to the ground or via some intermediate relay satellite. Laser satellite communication can play a pivotal role in the architecture of such systems.

Figures 1 and 2 show the data rates for earth observation satellites and the resolution of onboard optical or microwave image sensors.¹⁵ From these figures one can understand that the data-rate requirements for earth observation satellites are predicted to grow beyond 1 Gbps in the next

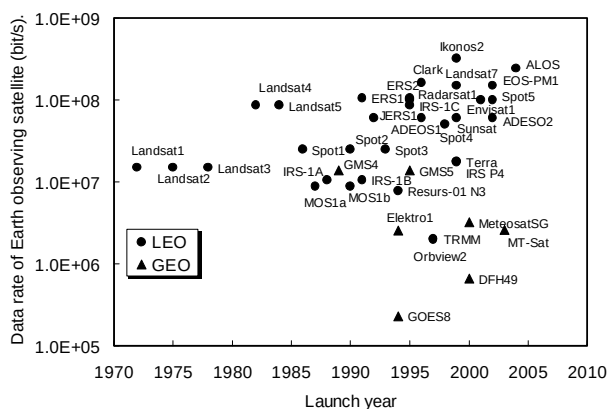


Fig. 1. Data rates for earth observation satellites.

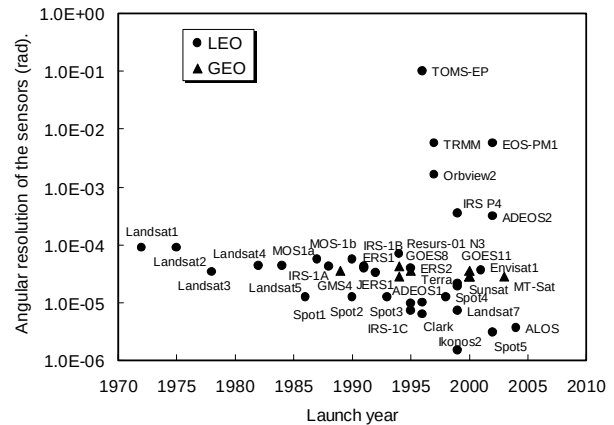


Fig. 2. Angular resolution of sensors on board earth observation satellites.

decade, and the angular resolution of the onboard sensors will be better than 1 μ rad. Lower vibrational disturbance platforms are needed to realize such high accuracy requirements, making it necessary to understand the characteristics of microvibration in order to ensure ultimate pointing accuracy.

3. Microvibrational disturbance

Recently, the characterization of the transfer functions of a structure with respect to launch vehicle environments has become a well recognized field with established data analysis techniques that are well developed. Nevertheless, when considering microvibrational levels problems must be considered that produce a wider frequency spectrum than encountered in classical structural modal testing. Even if the finite element model is fine-tuned after performing the classical structure modal test, many uncertainties remain with respect to microvibration of the satellite in orbit. The spacecraft attitude and orbit control system (AOCS) can compensate for low frequency disturbances (typically below 1 Hz), but is not able to control those at higher frequencies. These high frequency disturbances may have a significant impact on the spacecraft pointing stability (typically from 20 Hz up to 200 Hz), therefore we need to know their characteristics.

The microvibrations generated from mechanical moving devices on a satellite propagate along the structural panels and will disturb payloads or sensors requiring high pointing accuracy. A sinusoidal acceleration at the interface point of the payload in a satellite, which is due to microvibration generated from a mechanical moving source k and

sometimes induced and amplified by a structural resonance, is expressed by

$$a_k(t, f) = A_k(f)T_k(f)\exp\left(j2\pi ft - \frac{x_k}{d}\right), \quad (1)$$

where $A_k(f)$ is the amplitude of the microvibration at the frequency f on the disturbing device, $T_k(f)$ is the transfer function due to the structure, d is the average damping constant of the microvibration along the propagation path, and x_k is the distance from the disturbing source to the interface point. The variance of the acceleration at the frequency f is defined as

$$a_{k,rms}^2(f) = \overline{[a_k(t, f) - \overline{a_k(t, f)}]^2},$$

$$= \frac{A_k^2(f)T_k^2(f)\exp\left(-2\frac{x_k}{d}\right)}{2}. \quad (2)$$

The microvibrational level at the interface point generated from all the disturbing sources can be written as the sum of the variances because these quantities from all the sources are random variables independent of each other. Thus

$$a_{mount}^2(f) = \sum_{k=1}^N a_{k,rms}^2(f),$$

$$= \sum_{k=1}^N \left\{ \frac{A_k^2(f)T_k^2(f)}{2} e^{-2\frac{x_k}{d}} \right\}, \quad (3)$$

which corresponds to the PSD of the microvibration at the interface point of the payload, where N is the number of disturbing sources in the satellite. It is obvious that the term $T_k^2(f)\exp[-2x_k/d]$ is the transfer function from the vibrating source k .

In our situation, microvibrational displacement causes angular microvibration and the PSD of the angular microvibration is proportional to that of the displacement and is given by

$$\theta_{rms}(f) = \frac{C}{(2\pi f)^2} a_{mount}(f), \quad (4)$$

where C is a constant with unit of m^{-1} and $\theta_{rms}(f)$ is the angular disturbance at the mounting point of the laser communication terminal. The closed loop feedback gain $G(f)$ of the optical tracking servo system can reduce this angular disturbance. The residual tracking error of the laser communication terminal passing through the optical tracking servo system is expressed by

$$\theta_{error}(f) = \frac{\theta_{rms}(f)}{1 + G(f)},$$

$$= \frac{Ca_{mount}(f)}{4\pi^2 f^2 [1 + G(f)]}, \quad (5)$$

where $G(f)$ is the rejection gain of the angular perturbation of the optical tracking system at the interface point of the payload.

4. Experimental setup for microvibration tests

The main objectives of the system level microvibration test on the OICETS satellite are to identify 1) the effects of the dominant disturbance sources on the tracking performance, 2) the microvibrational characteristics of the disturbing sources, and 3) the transfer characteristics of microvibration on the satellite structure. These data will be used for comparison with in-orbit microvibration data.

The microvibration test was performed by using a simulated OICETS satellite model consisting of a model of the satellite structure and an engineering model of the LUCE. Measurements were made using 40 accelerometers with a sensitivity of 0.01 m/s^2 ($=1 \text{ mG}$) and a dynamic range of 50 dB.^{16,17} Some flight components, such as four reaction wheels (RWs), two conical-scanning earth sensors (CESs), one inertial reference unit (IRU), two solar paddle drive motors (PDMs), and one set of microvibration measurement equipment (MVE) were equipped with some dummy mass components. The structural panels are made of aluminum with sandwich honeycomb construction. The satellite bus structure has a height of 1.5 meters, a width of 0.8 m and a depth of 1.1 m. Figure 3 shows internal views of the OICETS satellite.

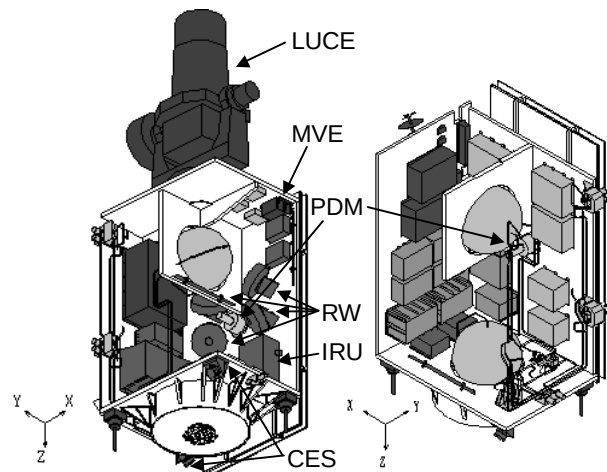


Fig. 3. Internal views of the OICETS. There are LUCE and MVE on the top panel, RWs and IRU on the $-Y$ panel, CESs on the base panel, and PDMs on $\pm Y$ panels.



Fig. 4. Configuration of the microvibration test in the ambient environment on the ground.

The satellite was suspended from a lifting tackle by four straps. A “quasi free-free” configuration was achieved for the satellite’s microvibration tests by using low frequency suspension devices such as a soft sling suspension. Such devices also isolate the test specimen from external mechanical disturbances. Figure 4 shows the configuration of the microvibration test. The complete system was installed in a booth surrounded by a blackout curtain, and the air-conditioning was turned off during measurement sequences. Most microvibration tests were performed at night. These arrangements ensured a reduction of laminar airflows, external mechanical and acoustical noise.

The test specimen was very close to the in-orbit configuration, except there were no solar arrays, empty tanks, or insulation. Differences to in-orbit operating conditions were: the in-orbit operational mode of AOCS was not initiated but all mechanisms were turned on; there was no thermal environment control; no radiation effects on the electronics; and the structure was in ground-level ambient and gravity environments.

5. Results of microvibration tests

5.1. Acceleration measurements

Reaction wheels

Figure 5 shows the acceleration of reaction wheel No. 4 (RW4) as a function of the wheel rotation speed. The acceleration level from RW4 changes with the

wheel rotation speed and depends on the degree of imbalance between the rotating forces of the wheel and structural resonance modes of its housing and support structure. The wheel rotation speed is fixed at the highest acceleration level in the microvibration test when all the disturbing devices are turned on.

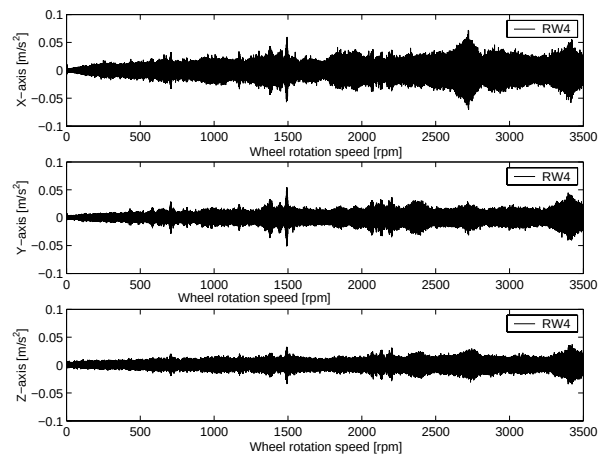
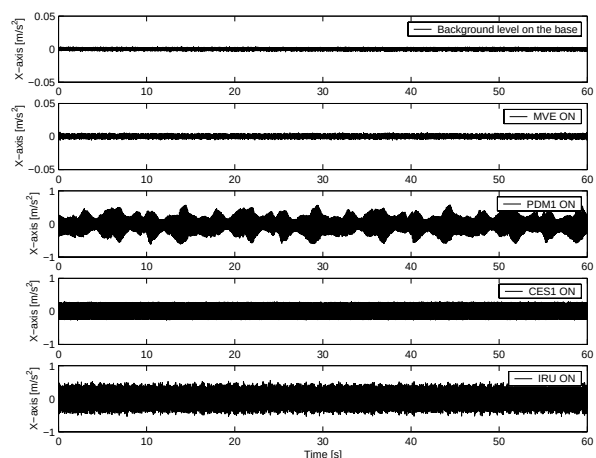


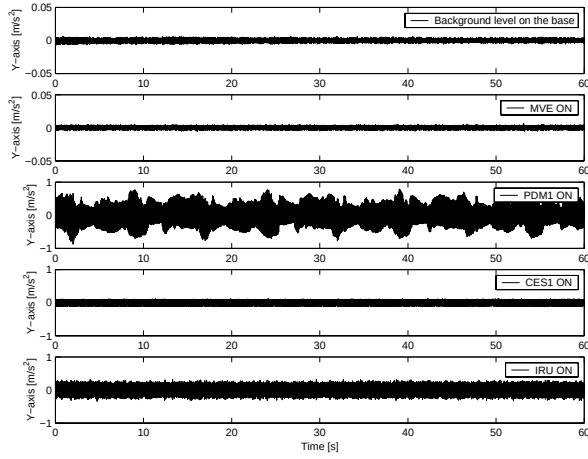
Fig. 5. Acceleration responses on RW4 as a function of the wheel rotation speed.

Other devices

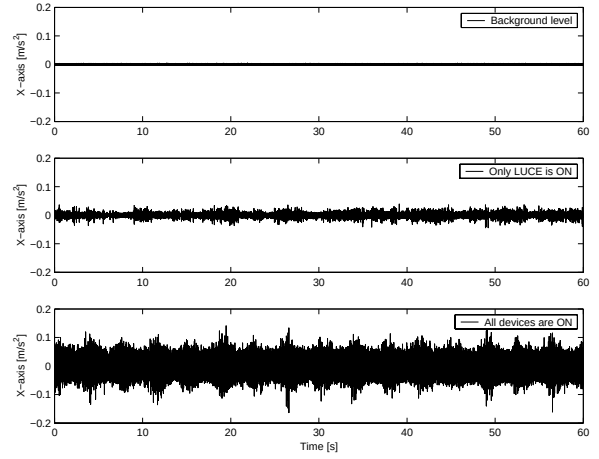
Figures 6(a), (b) and (c) show the typical acceleration responses along the X, Y and Z axes for each mechanical moving device such as MVE, PDM No. 1 (PDM1), CES No. 1 (CES1) and IRU in the satellite, which have been measured separately. The acceleration signatures for each device are different. The maximum acceleration level of PDM1 is 0.8 m/s^2 ($\approx 80 \text{ mG}$) which is 40 dB higher than that of the background level.



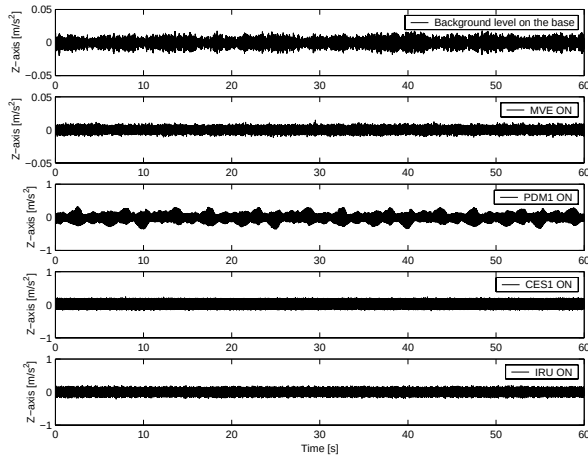
(a) X-axis acceleration responses



(b) Y-axis acceleration responses



(a) X-axis acceleration responses

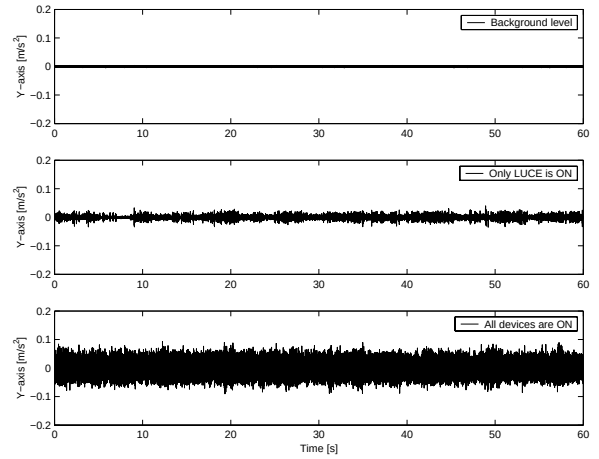


(c) Z-axis acceleration responses

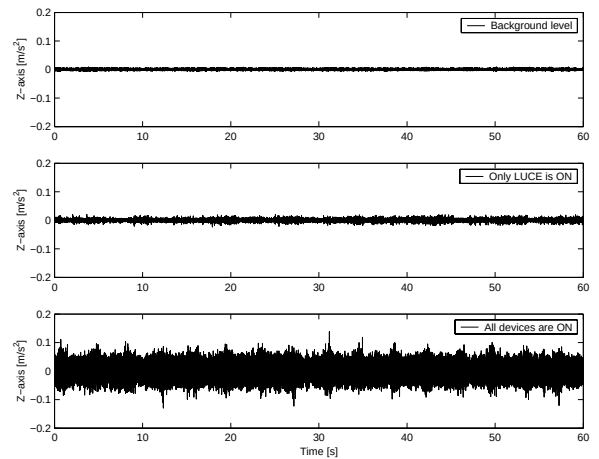
Fig. 6. Typical acceleration responses along the X, Y and Z axes for the satellite support base used during ground testing and the MVE, PDM1, CES1 and IRU devices in the satellite, each of which is turned on and measured separately.

Interface point of the payload

Figures 7(a), (b) and (c) show the acceleration levels of the mounting point of the laser communication terminal along the X, Y and Z axes, respectively. The acceleration of about 0.1 m/s^2 ($\approx 10 \text{ mG}$) of the interface point was measured as the worst case when all the disturbing devices in the satellite were turned on. Of all the noise sources, RW4 was the largest contributor to vibration at the mounting point of the laser communication terminal. Even the laser communication terminal itself becomes one of the noise sources, as can be observed from Fig. 7 when only the LUCE is turned on. It is the gimbal drive motors of the LUCE that generate the microvibrational disturbance.



(b) Y-axis acceleration responses



(c) Z-axis acceleration responses

Fig. 7. Typical acceleration levels of the interface point of the payload when all the disturbing devices are turned on while the LUCE is tracking a target.

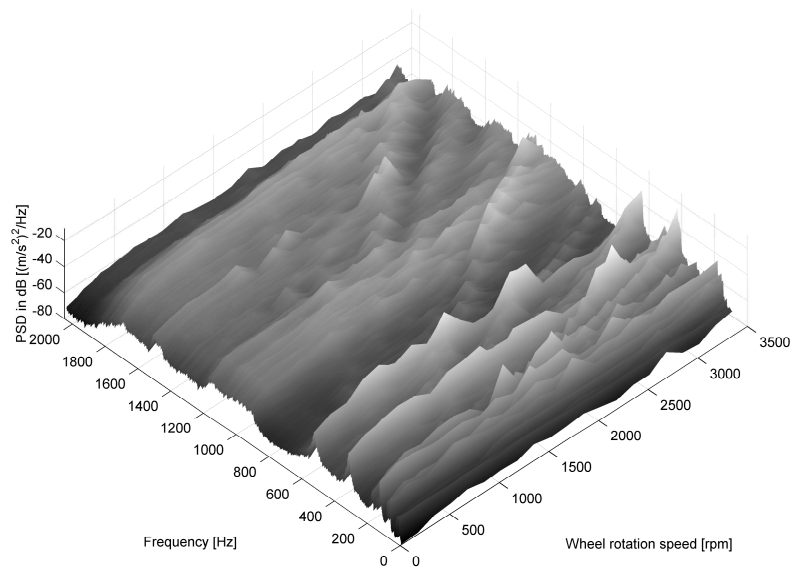


Fig.8. RW4 frequency response as a function of rotation speed.

5.2. Power spectral densities

Reaction wheels

The PSDs of the acceleration levels of RW4 are shown in Fig. 8 when the wheel rotation speed is swept over a range of values. There are two contributions to the frequency components. One comes from the unbalanced rotating forces of the wheel which produces frequencies proportional to the wheel rotation speed, and some harmonics. The other is caused by the structural resonance modes of RW4's housing and support structure, which has several fixed resonance frequencies.

Other devices

Figure 9 shows the typical PSDs of MVE, PDM1, CES1 and IRU. The X, Y and Z accelerations are added as root sum squares (rss). The PSD of the MVE remains at the same level as the background noise when the MVE is turned on, so it does not disturb the other devices. The PDM1 has a wide frequency spectrum produced by the stepping motor. The CES1 generates harmonics in the spectrum caused by the scanning sensor head rotating at a frequency of 2 kHz. The IRU produces the dominant frequency spectrum generated by the gyro wheels.

Interface point of the payload

The PSDs of the accelerations at the mounting point of the laser communication terminal are shown in Fig.

10, both for when all the disturbing devices in the satellite are turned on and, when only the LUCE is turned on to track a target. The higher frequency components at the interface point come from the mechanical moving devices. The LUCE itself also generates the lower frequency components below several tens of Hertz along the X-axis, which is the elevation axis of the gimbal motor rotation of the laser communication terminal.

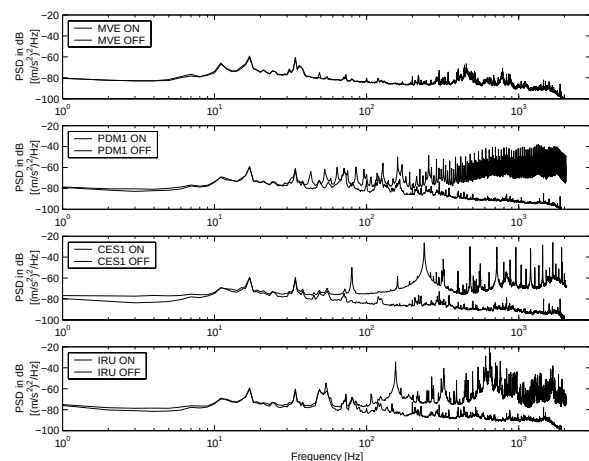


Fig. 9. Typical PSDs on MVE, PDM1, CES1 and IRU when each sensor is turned on and off separately.

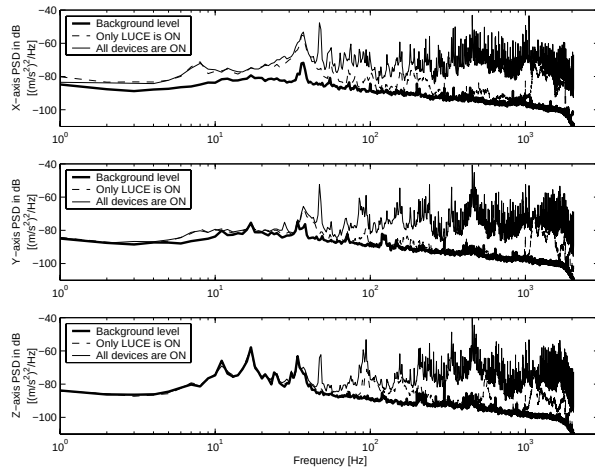


Fig. 10. PSDs of the accelerations at the interface point of the payload when all the disturbing devices are turned on and when only the LUCE is turned on.

5.3. Damping characteristics

Damping constant

The microvibrational level is damped as a function of the distance from the vibrating source. The damping characteristics of the acceleration are shown in Fig. 11. The damping constant d is given experimentally by

$$d = \frac{1}{1.7434} = 0.574, [\text{m}] \quad (6)$$

which is the $1/e$ damping length of the amplitude of the acceleration. One can predict the acceleration level at any point by using Eqs. (3) and (6), and so give proper consideration to the effects of microvibrations in the preliminary design phase of the satellite.

Figure 12 shows the PSDs of the accelerations at several of the sensor positions when the noise source is only RW4. The frequency components of the accelerations above about 100 Hz are reduced, however, the lower frequencies below several tens of Hertz are not damped, that is, they propagate along the satellite panels or structures unattenuated. The lower frequency region is mainly due to bending and twisting vibrational mode resonances of the structure that can be predicted by the classical structural modal analysis of the satellite. The OICETS satellite has a natural frequency of 25 Hz as the first mode of the structure when in-orbit.

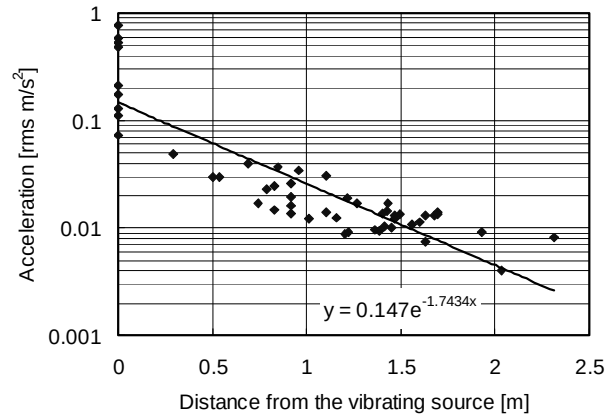


Fig. 11. Damped acceleration as a function of the distance from the disturbing sources to the measured positions.

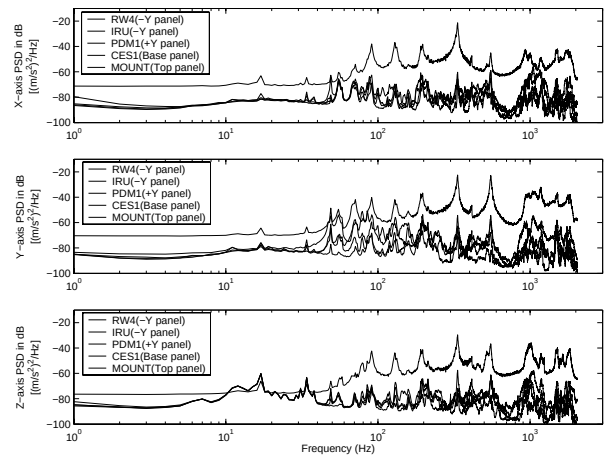


Fig. 12. PSDs of the microvibrations at each sensor position of RW4, IRU, PDM1, CES1 and the mounting point when the noise source is only RW4.

5.4. Transfer functions

Disturbing sources to the mounting point

Figure 13 shows the transfer functions of the microvibrations when the disturbing source is only RW4, as shown in Fig. 12. That is, Figure 13 shows the effect of microvibrations propagating along the satellite structural panels from RW4. The structure itself acts as a low pass filter and some frequency components at lower frequencies below several tens of Hertz can propagate without damping. On the other hand, the higher frequency components above a hundred Hertz reduce with distance from the disturbing sources.

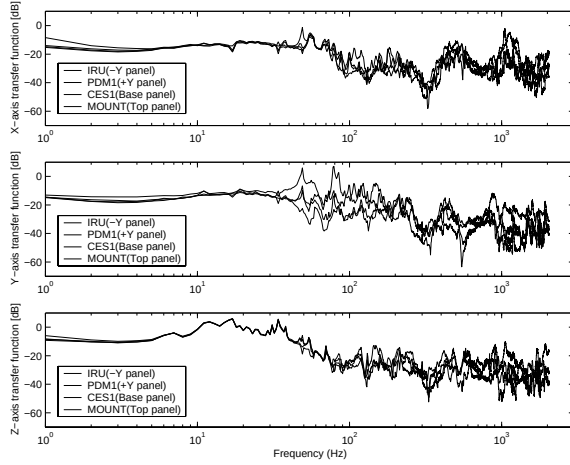


Fig. 13. Transfer functions of the microvibrations at each sensor position of IRU, PDM1, CES1 and the mounting point when the disturbing source is only RW4.

Mounting point to the optical antenna

Figure 14 shows the transfer functions of the microvibrations from the interface point of the spacecraft to the LUCE optical antenna both when LUCE is operational and when it is turned off. LUCE's structure acts as a low pass filter with a cut-off frequency of about 100 Hz, and higher frequency components of the microvibration are also reduced while propagating along the structure. On the other hand, the lower frequency components seem to be amplified since LUCE generates low frequency

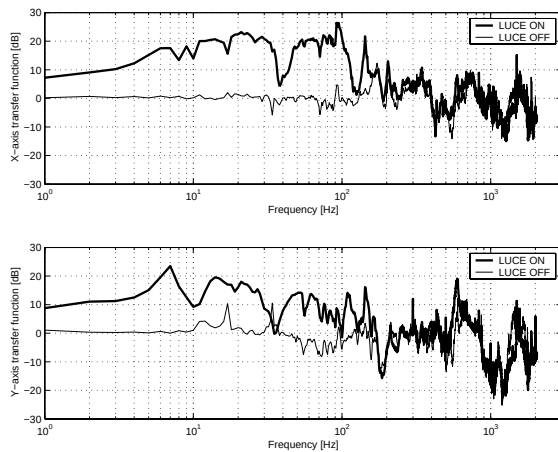


Fig. 14. Transfer functions of the microvibrations from the mounting point to the LUCE optical antenna while LUCE is tracking a target, and when it is turned off. The only disturbing source is RW4.

disturbances from its gimbal drive motors as shown in Fig. 10. When designing high accuracy tracking systems, engineers should take into account the disturbances from not only the spacecraft base motion but also the tracking mechanism itself.

6. Comparison with laser tracking performances

PSD of the angular tracking error

The residual angular tracking errors of the laser communication terminal can be obtained using

$$\begin{cases} \theta_{error,x}(f) = \frac{C_x a_{mount,x}(f)}{4\pi^2 f^2 [1+G(f)]}, \\ \theta_{error,y}(f) = \frac{C_y a_{mount,y}(f)}{4\pi^2 f^2 [1+G(f)]} \end{cases} \quad (7)$$

which have the unit of $[\text{rad}/\text{Hz}^{1/2}]$. $a_{mount,x}(f)$ and $a_{mount,y}(f)$ correspond to the microaccelerations on the interface point that influence the x and y tracking directions, respectively. The constants C_x and C_y are experimentally determined to be

$$\begin{cases} C_x = 51.96 \quad [\text{m}^{-1}] \\ C_y = 13.32 \quad [\text{m}^{-1}] \end{cases} \quad (8)$$

Figure 15 shows the PSDs of the angular tracking errors of the laser communication terminal measured at the target terminal simulator. For the comparison, the PSD's of the tracking errors due to the microaccelerations on the mounting point of the laser communication terminal are estimated from Eq. (7).

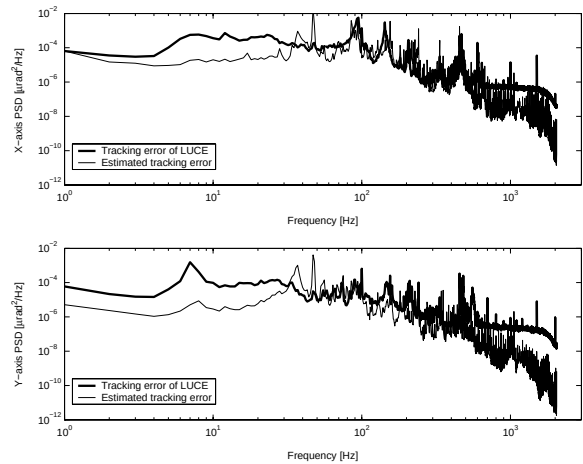


Fig. 15. PSDs of the angular tracking errors. The optical tracking errors of the laser communication terminal measured at the target terminal simulator are indicated by the bold line and the estimated tracking errors from Eq. (7) are also shown.

The low frequency components come from mainly the gimbal drive motors of the laser communication terminal itself, which can be seen also from Fig. 10. At higher frequencies the disturbances mainly come from the mechanical moving devices in the satellite bus system, and the electrical noise of the tracking sensor which is the dominant contributor to the tracking errors at higher frequencies.

Tracking sensor noise

Figure 16 shows the comparison of the PSD of the angular tracking errors when different irradiances impinge on the tracking sensors. When the received optical power is a minimum, the noise spectra increase because the sensor noise dominates in the tracking signal. In order to suppress the frequency spectra of the angular tracking errors, the sufficient optical signal on the tracking sensor is needed for not only the required optical power for the communication but also the residual tracking error.

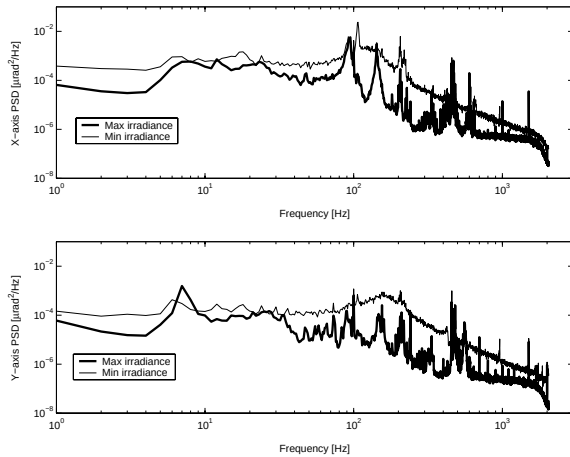


Fig. 16. Comparison of the PSDs of the angular tracking errors when the different irradiances impinge on the tracking sensor.

Angular disturbance on the interface point

As Eq. (4) gives the relation between the angular disturbance and the acceleration level, we can estimate the PSD of the angular disturbance at the mounting point of the laser communication terminal and the result is shown in Fig. 17. The rms value of the spacecraft base motion at the interface point of the payload is given by

$$\Phi_{rms} = \sqrt{\int \theta_{rms}^2(f) df} = 85.33 \text{ [μrad rms]}, \quad (9)$$

which is estimated from the PSD from 1 to 2048 Hz

in Fig. 17 and this is the worst case condition that includes microvibration from LUCE itself. The worst-case spacecraft vibrational motion, without the disturbance from LUCE is estimated to be 74.78 μrad rms from Eq. (4). These spectra are considered to be reasonable levels compared with other published data.^{4-6,12,17}

The in-orbit microvibrations may be greater because during the ground tests, part of the microvibrational energy is dissipated into the air as heat or sound, which does not occur in the space environment. On the other hand, as there is no acoustical noise in orbit, this contribution becomes negligible. These effects must be verified with in-orbit tests in the future.

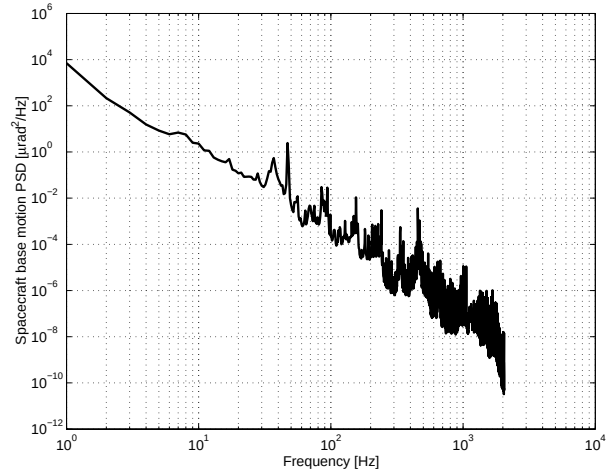


Fig. 17. PSD of the worst case angular microvibrational disturbance of the OICETS satellite at the interface point of the payload estimated from the microaccelerations.

7. Conclusion

Microvibrations on the OICETS satellite have been measured and analyzed using data acquired during tests on the ground. The acceleration level on the mechanical moving devices reached about 0.8 m/s² (=80 mG). At the interface point of the laser communications payload, the damped acceleration level was measured to be about 0.1 m/s² (=10 mG) when all the disturbing sources are turned on in the satellite. The rms value of the angular microvibrational disturbance on the spacecraft, including the microvibrational disturbance from the payload itself, is estimated to be 85 μrad rms, which is the worst case disturbance on the OICETS satellite. Optical tracking by the laser communication terminal was maintained even in the worst case condition,

which adequately satisfies the requirements for the optical tracking performance required for intersatellite laser communication links.

8. Acknowledgments

The authors would like to express their appreciation to Mr. Toshihiko Kurii and Mr. Koichi Shiratama for acquiring the data in the microvibration tests, as well as all the members of the OICETS project group at NEC TOSHIBA Space Systems, Ltd., as the prime contractor of the LUCE and the OICETS satellite system.

9. References

- [1] B. Dekel, S. Israelian, M. Garber, G. Birin, E. Zamir, "Three-axis star tracker," in Optical Engineering and Remote Sensing, M. Oron, I. Shladov, and Y. Weissman, eds., Proc. SPIE 1971, 203-211 (1993).
- [2] L. W. Cassidy and L. Schlom, "Miniature star tracker for small remote sensing satellites," in Platforms and Systems, W. L. Barnes and B. J. Horais, eds., Proc. SPIE 2317, 160-166, (1995).
- [3] K. Noguchi, K. Sato, R. Kasikawa, N. Ogura, K. Ninomiya, T. Hasimoto, E. Hirokawa, "CCD star tracker for attitude determination and control of a satellite for a space VLBI mission," in Space Sciencecraft Control and Tracking in the New Millennium, E. K. Casani, M. A. V. Does, eds., Proc. SPIE 2810, 190-200 (1996).
- [4] J. Sudey and J.R. Sculman, "In orbit measurements of Landsat-4 thematic mapper dynamic disturbances", 35th Congress of the International Astronautical Federation, Lausanne, Switzerland, IAF-84-117 (1984).
- [5] K. J. Held and J. D. Barry, "Precision pointing and tracking between satellite-borne optical systems," Optical Engineering, Vol. 27, No. 4, 325-333 (1988).
- [6] M. Wittig, K. van Holts, D. E. L. Tunbridge, and H. C. Vermeulen, "In-orbit measurement of microaccelerations of ESA's communication satellite OLYMPUS," Proc. SPIE, Vol. 1218, 205-214 (1990).
- [5] V. Guillaud, C. Champetier, A. Capitaine, Y. M. Lefevre, "Mastering microvibrations for high pointing accuracy systems," Proc. of 2nd ESA International Conference on Spacecraft GNC Systems (1994).
- [7] K. I. Schultz and S. Fisher, "Ground-based laser radar measurements of satellite vibrations," Applied Optics, Vol. 31, No. 36, 7690-7694 (1992).
- [8] K. I. Schultz, D. G. Kocher, J. A. Daley, J. R. Theriault, J. Spinks, and S. Fisher, "Satellite vibration measurements with an autodyne CO2 laser radar," Applied Optics, Vol. 33, No. 12, 2349-2355 (1994).
- [9] E. M. Marchante, L. Munoz, "ARTEMIS satellite microvibrations testing and analysis activities," 48th IAC, Torino, Italia, October (1997).
- [10] C. Caleazzi, P. C. Marucchi-Chierro, L. V. Holtz, "Experimental activities on ARTEMIS for the microvibration verification," Proc. of ESA's conference on Spacecraft Structures, Materials and Mechanical Testing in Noordwijk, The Netherlands, 27-29 March (1996).
- [11] M. Toyoshima, K. Araki, Y. Arimoto, M. Toyoda, M. Jeganathan, K. Wilson, and J. Lesh, "Reduction of ETS-VI Laser Communication Equipment Optical-Downlink Telemetry Collected During GOLD," The Telecommunications and Data Acquisition Progress Report 42-128, Jet Propulsion Laboratory, 1-9 (1997).
- [12] M. Toyoshima and K. Araki, "In-orbit measurements of short term attitude and vibrational environment on the Engineering Test Satellite VI using laser communication equipment," Optical Engineering, Vol. 40, No. 5, pp. 827-832 (2001).
- [13] S. Arnon and N. S. Kopeika, "Laser Satellite Communication Network—Vibration Effect and Possible Solutions," Proceedings of the IEEE, Vol. 85, No. 10, pp.1646-1661, October (1997).
- [14] M. Privat, "On ground and in orbit microvibrations measurement comparison," Proc. of the 8th European Space Mechanisms and Tribology Symposium, ESA SP-438 (1999).
- [15] D. Baker, ed., Jane's Space directory, Sixteenth Ed. 2000-2001, Jane's Information Group, Surrey (2000).
- [16] T. Kurii, K. Shiratama, and K. Nakagawa, "Pointing control system for Optical Intersatellite Link Terminal on-board OICETS," the 45th Space Science and Technology Conference 2001, JSASS.
- [17] T. Jono, M. Toyoshima, N. Takahashi, T. Yamawaki, K. Nakagawa and A. Yamamoto, "Laser tracking test under satellite microvibrational disturbances by OICETS ATP system," in Acquisition, Tracking, and Pointing XVI, M. K. Masten; L. A. Stockum; eds., Proc. SPIE 4714, 97-104 (2002).